



GEETHANJALI COLLEGE OF
ENGINEERING & TECHNOLOGY

...Striving Towards Perfection

Eco-Verify: A Comparative Gap Analysis Tool for Corporate and Consumer Social Responsibility Using Machine Learning-Based Topic Modeling and NLP

Submitted by : Haripriya Dasaram

Roll no: 25R11D5801

Department of computer science

Under the guidance of Dr.K.Kamakshaiah

ABSTRACT

- Consumer Social Responsibility (CnSR) refers to the ethical, environmental, and social responsibilities demonstrated by consumers through their purchasing decisions and online opinions. With the rapid growth of social media and e-commerce platforms, consumers generate massive amounts of textual data in the form of reviews, comments, and discussions. This project utilizes **Natural Language Processing (NLP)** and **Machine Learning-based Topic Modeling** to analyze consumer opinions and identify major themes related to social responsibility. The collected text data undergoes preprocessing techniques such as tokenization, stop-word removal, stemming, and lemmatization. Topic modeling algorithms such as **Latent Dirichlet Allocation (LDA)** or **Non-negative Matrix Factorization (NMF)** are then applied to discover hidden topics. The project helps organizations understand consumer expectations regarding sustainability, ethical products, environmental concerns, and corporate responsibility. The system provides visualizations of extracted topics and assists decision-makers in developing socially responsible business strategies.

INTRODUCTION

Consumers today are increasingly concerned about:

- Environmental sustainability
- Ethical sourcing
- Fair labor practices
- Eco-friendly packaging
- Corporate transparency
- Millions of reviews and social media posts contain valuable insights regarding these concerns. Manual analysis is difficult due to the huge volume of data.
- Natural Language Processing (NLP) and Machine Learning can automatically analyze these texts and identify common themes using **Topic Modeling**.

PROBLEM STATEMENT

Organizations struggle to understand consumer perspectives on social responsibility because:

- Large amounts of unstructured text exist
- Manual analysis is time-consuming
- Hidden themes cannot be easily identified
- Consumer expectations change rapidly
- Therefore, an automated topic modeling system is required.



OBJECTIVES

- Collect consumer reviews or social media comments.
- Preprocess textual data using NLP techniques.
- Apply Machine Learning-based Topic Modeling.
- Discover major Consumer Social Responsibility themes.
- Visualize extracted topics.
- Help organizations understand consumer concerns.



EXISTING SYSTEM

Current Approach

- Manual review analysis
- Basic sentiment analysis
- Keyword searching
- Surveys

Limitations

- Time-consuming
- Human bias
- Difficult for large datasets
- Hidden topics remain undiscovered

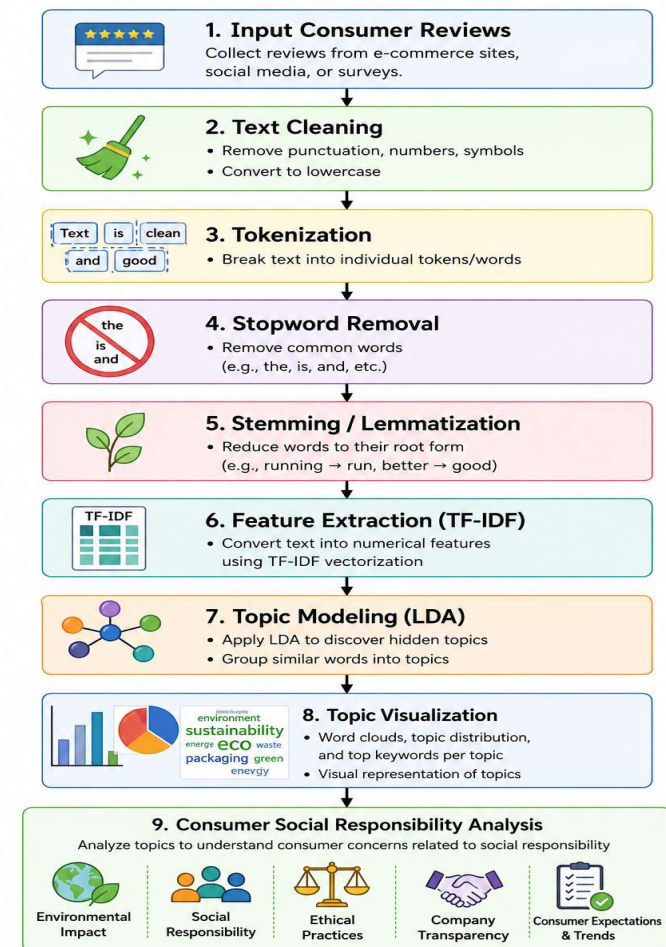
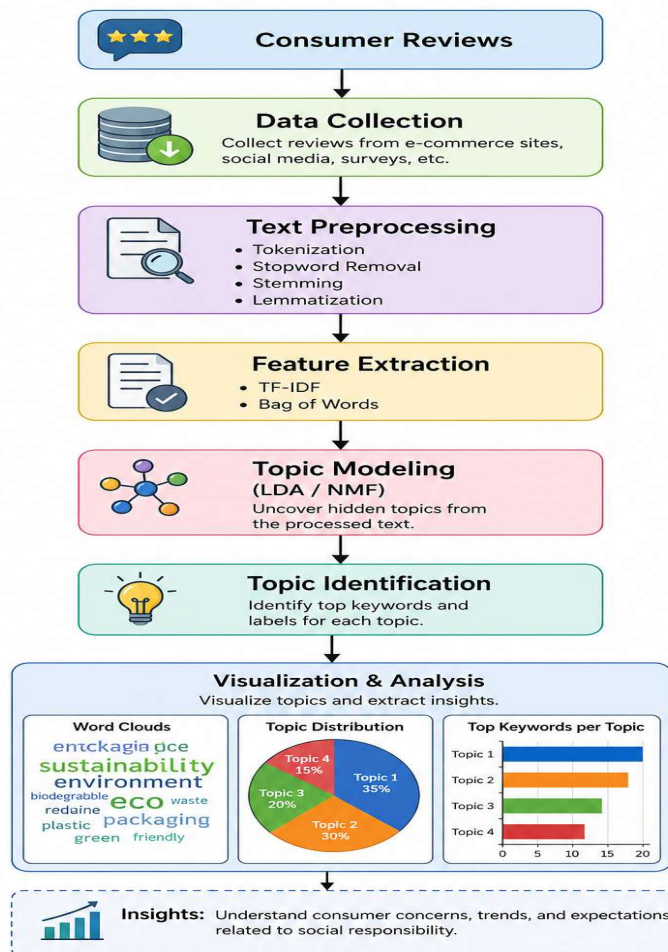


PROPOSED SYSTEM

The proposed system:

- Collects textual consumer data
- Cleans and preprocesses text
- Converts text into numerical features
- Applies Topic Modeling (LDA/NMF)
- Generates meaningful topics
- Visualizes results using graphs and word clouds

SYSTEM ARCHITECTURE AND WORKFLOW





ALGORITHMS USED

Natural Language Processing

- Tokenization
- Stopword Removal
- Stemming
- Lemmatization

Machine Learning

- TF-IDF Vectorizer
- LDA
- NMF



SOFTWARE REQUIREMENTS

Requirement

Operating System

Programming Language

IDE

Libraries

Database

Specification

Windows 10/11

Python 3.10+

VS Code/Jupyter Notebook

NLTK, Scikit-learn, Pandas, NumPy,
Matplotlib, WordCloud

CSV or Excel



HARDWARE REQUIREMENTS

Component

Requirement

Processor

Intel i3 or above

RAM

4 GB

Storage

10 GB free space

Internet

Required for dataset collection

Display

1366×768 resolution



ADVANTAGES

- Automatic topic discovery
- Handles large datasets
- Faster than manual analysis
- Supports business decision-making
- Identifies emerging consumer concerns
- Easy to visualize insights



EXPECTED OUTCOME

The system will:

Analyze consumer text data.

Extract major Consumer Social Responsibility topics.

Display keywords for each topic.

Generate visualizations such as word clouds and charts.

Provide insights that organizations can use to improve socially responsible practices.



CONCLUSION

This mini project demonstrates how **Machine Learning** and **Natural Language Processing** can be combined to analyze Consumer Social Responsibility from large collections of text. By applying **topic modeling**, the system uncovers hidden themes related to sustainability, ethics, and consumer expectations, enabling organizations to make more informed and socially responsible decisions.



REFERENCES

Speech and Language Processing – Daniel Jurafsky & James H. Martin.

- Introduction to Information Retrieval – Christopher D. Manning, Prabhakar Raghavan & Hinrich Schütze.
- NLTK documentation.
- scikit-learn documentation.
- Research papers on Topic Modeling using LDA and NLP in consumer behavior analysis.



THANK YOU!